Understanding data merging

by Shahriar Analytical Academy

What Is Data Merging?

1 Definition

Data merging is the process of **combining two or more datasets** into a single, unified dataset based on a common column or key. It is also called **joining** data.

2 Purpose

Bring together data from different sources

Create a richer dataset with more information

Prepare for comprehensive analysis and reporting

3 Real-Life Analogy

Think of two puzzle pieces. One has names and the other has phone numbers. Merging joins them using a common identifier — like name — to create a full picture.

Why Merging Data Is Important

1

Combine Multiple Perspectives

Link customer orders with customer details, or employee IDs with performance records.

2

Reduce Redundancy

Keep related information in separate tables and merge them when needed.

3

Prepare for Analysis

Many insights require combined data (e.g., joining sales with region to analyze sales by location).

4

Integrate Data from Various Systems

Merge data from different tools like CRM, ERP, or survey platforms into one analytical table.

Key Terms to Understand Before Merging

Key Terms

- **Key/Join Column:** The column that exists in both tables and is used to match rows (e.g., customer_id).
- **Primary Table (Left):** The main dataset you start with.
- **Secondary Table (Right):** The dataset you're merging in.

Types of Merge Relationships

- **One-to-One:** One match in both tables
- **One-to-Many:** One value in one table matches multiple rows in the other
- **Many-to-Many:** Multiple matches on both sides (requires caution)

Merge Types (Joins) Explained

Join Type	Description	Result
Inner	Keeps only matching rows from both tables	Common records only
Left	Keeps all rows from left table, and matches from right	All from left, fill NaNs if no match in right
Right	Keeps all rows from right table, and matches from left	All from right, fill NaNs if no match in left
Outer	Keeps all rows from both tables, fills missing with NaN	Full combination, shows unmatched from both sides

Common Pitfalls and How to Avoid Them

Mismatched Column Names

Ensure join keys are spelled identically or use left_on and right_on in pandas.

Duplicates in Keys

Multiple matches can cause bloated or inaccurate merges — check for duplicates.

Null Values in Join Columns

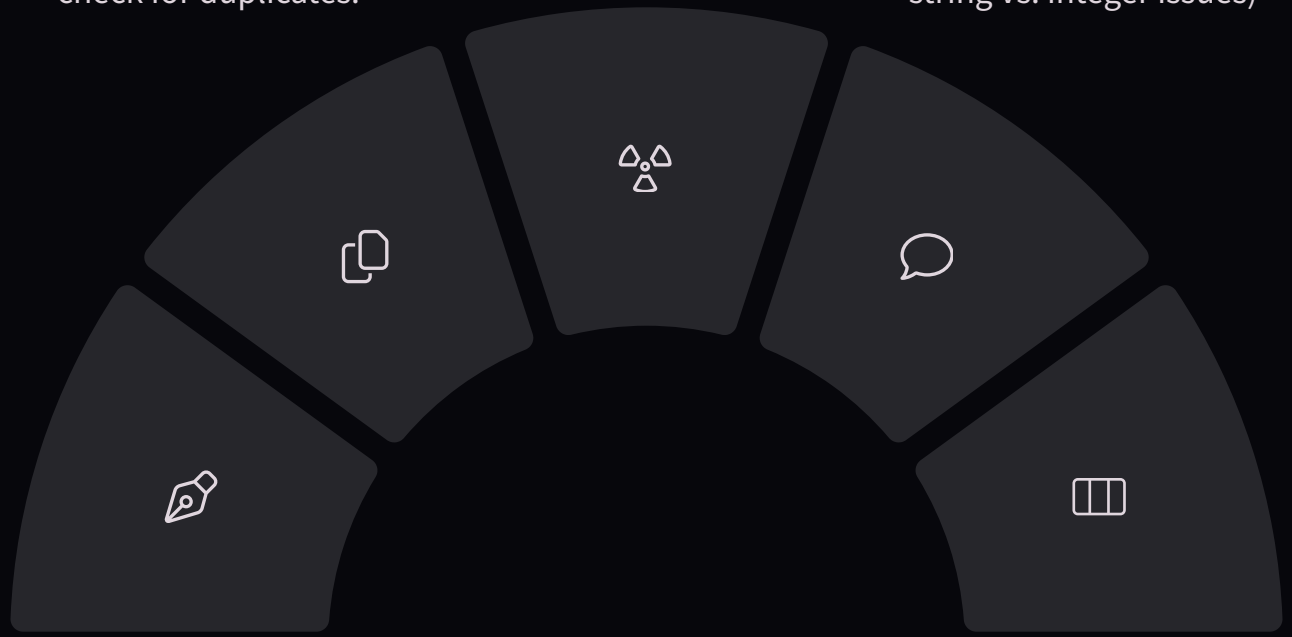
Merges won't match rows with NaN keys unless handled.

Mismatched Data Types

Join columns must be the same data type (e.g., string vs. integer issues)

Overlapping Column Names

Columns with the same name but different values will be auto-renamed — use suffixes.



Summary – Key Learning Points

1

Data merging is the process of combining datasets based on shared keys

2

It is crucial for analysis that requires data from multiple tables or sources

3

Merge types (inner, left, right, outer) control how rows are matched and preserved

4

Watch out for pitfalls like null keys, mismatches, and column name conflicts

5

Clean, validate, and document your merge process for reliable results

Thank You!

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